

When logic wins over machine learning

When we drive, we are faced with unexpected potential dangers—people crossing the road, dogs that don't obey their masters, children riding cycles and more. However, from experience we learn to ignore some and pay more attention to others. But most autonomous cars try to handle each situation theoretically, which is too tiresome for the car and its occupants!

nuTonomy, a company that has been testing its robo-taxis in Singapore's One-North technology business district, uses formal logic to handle such complex situations. Formal logic is based on a hierarchy of rules, with priorities. For example, the rule "don't hit pedestrians" has a greater priority than "don't hit other vehicles," which is more important than "don't hit objects." The car attempts to apply all the rules, but in the given order of priority. It uses a planning algorithm called RRT* to evaluate many potential paths based on the data from cameras and other sensors. Decision-making software evaluates these paths and selects the one that fits in with the rule hierarchy.

The idea of formal logic is different from machine learning. In machine learning, you show the algorithm innumerable driving scenarios, from which it figures out the rules of good driving and then applies them to real-life situations. nuTonomy uses machine learning to interpret sensor data but not for decision-making because the company feels that machine learning is like a 'black box' and it is difficult to figure out why machine-learning systems make the choices



nuTonomy relies on formal logic for their robo-taxis to make the right decisions (Courtesy: nuTonomy)

they do. Formal logic, on the other hand, gives more control over the machine and guarantees that the car will obey rules in the order of importance and stay safe even in unprecedented situations. "It's a rigorous algorithmic process that's translating specifications on how the car should behave into verifiable software," explains nuTonomy CEO and cofounder Karl Iganemma, in a press report. "That's something that's really been lacking in the industry."

timings, the Surtrac system applies AI algorithms to data collected by the radar sensors and cameras of computerised traffic lights to dynamically build a timing plan. The system is decentralised and each signal makes its own timing decision. It also sends the data to traffic intersections downstream so they can plan ahead.

There are 50 such smart intersections now, with plans for expansion citywide. Following that, Smith's group wants to improve the system to enable signals to talk to cars! According to an IEEE news report, they have already installed short-range radios at 24 intersections. Such systems are expected to begin being built into some cars this year. Traffic signals can then let drivers know of upcoming traffic conditions or change in lights, increasing safety and relieving congestion. The vehicle-to-infrastructure communication system could also prioritise certain vehicles like public transport buses.

AI is helpful on social media too. Facebook, for instance,

uses AI to spot and remove offensive content. It is also planning to integrate AI-based suicide prevention tools into Facebook Live and Messenger, in order to recognise and help people with suicidal tendencies.



Bosch-backed robot Kuri (Courtesy: Mayfield Robotics)

In April this year, Mark Zuckerberg unveiled a platform that transforms users' smartphone camera into an engine for augmented reality (AR). The solution relies on implementing AI on the network edge. The platform lets users to layer digital effects atop images and videos captured by the camera. One of the fun demos showed digital sharks swimming around a bowl of cereal.

Facebook has more real-world plans for the future. For example, you can pin a virtual note on your fridge and your roommates will be able to see it when they view the fridge through their cameras.

Neural networks help to identify people and track their movement and activities within the camera's FOV, in order to apply appropriate digital effects. Facebook's deep neural networks run on the phone itself because getting across the Internet and back will be too slow to effectively implement such digital effects.

While Facebook has optimised its deep learning technology to run on current-day mobile phones,